

Simulating a 1DOF vibrating system using Simscape

0 Intro

0.1 What is Simulink?

Simulink is a graphical programming environment for modelling, simulating and analysing dynamic systems. Its Graphical User Interface (GUI) shows a block diagram representing the system you are modelling. Sub-functions appear as blocks, and most built-in Matlab functions pre-exist in libraries.

Simulation is in the time-domain and connections from block to block carry signals. These can be sampled data, but by default they are a more sophisticated signal representation that allows Simulink to solve for system behaviour versus time. This can be sampled for easier analysis. A model runs from a start time (usually 0 seconds) to a stop time (editable and shown in the default view).

Simulink offers tights tight integration with the rest of the MATLAB environment:

- You can read and write variables in the Matlab workspace from Simulink
- You can call Matlab functions from Simulink
- You can run Simulink models from Matlab

Some tips for navigating Simulink:

- Double click on a block to edit its settings
- Use the mouse wheel to zoom in and out
- Hold the mouse wheel and move the mouse to change your view

0.2 What is Simscape?

Simscape is version of Simulink that makes the block diagrams look more like the mechanical or electrical system diagrams we might sketch on paper. In mechanical systems you'll see components such as masses, springs and dampers. In electrical systems you'll see components such as resistors capacitors and inductors. But there are many more sophisticated components included too. Connections now usually carry two quantities e.g. voltage and current, or position and velocity. The solver tries to make these all balance, based on the network of components that relate them.

You can see Simscape and Simulink versions of the system we will study compared at https://uk.mathworks.com/help/physmod/simscape/ug/mass-spring-damper-in-simulink-and-simscape.html. You'll notice that the Simulink version mentions "s" in various place, analogous to how one solves a differential equation by hand using an *Auxiliary Equation* (i.e. a Laplace transform). The Simscape model in contrast looks more like a diagram of the physical system and is easier to understand.

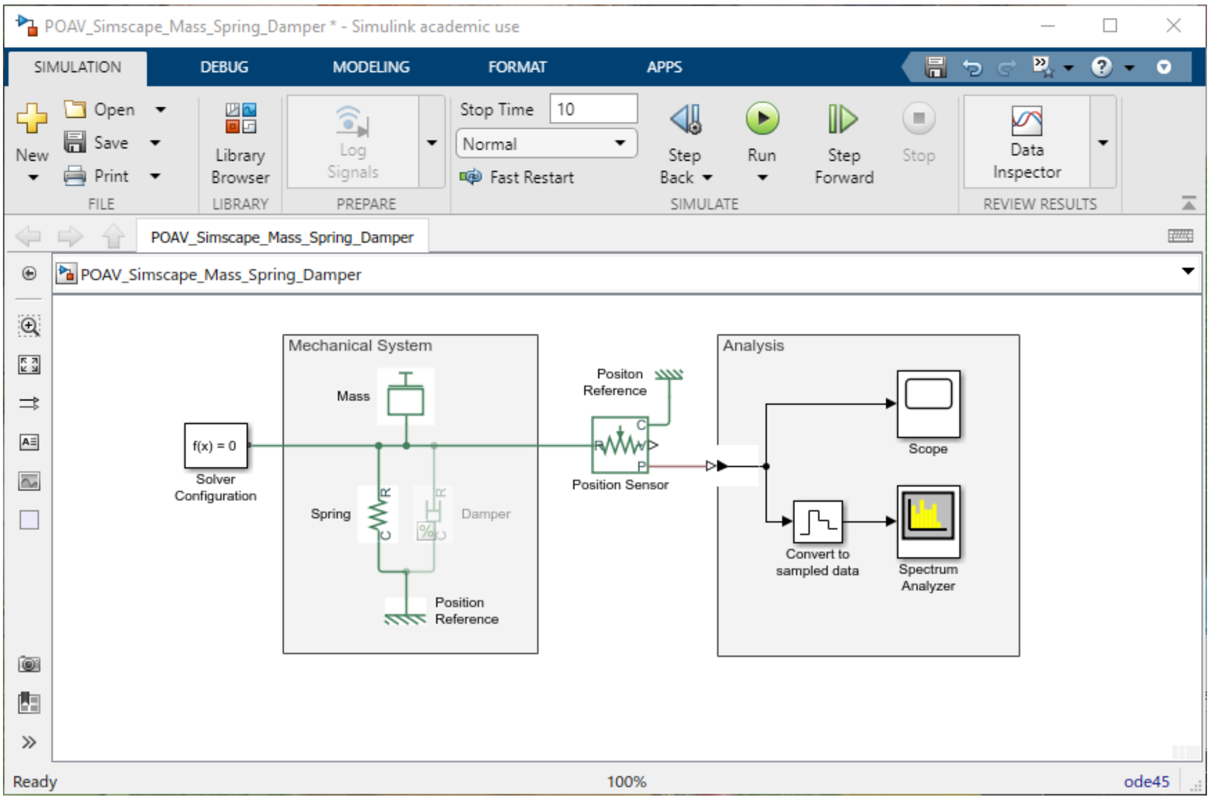
0.3 Getting Simscape

If you don't already have Simscape installed with Matlab then you'll need to add it. To do this:

- On the Matlab *Home* tab click on *Add-Ons* then *Get Add-Ons.*
- In the *Add-On Explorer* window that opens, type "Simscape" in the search board and press *Enter*.
- Simscape should be the top hit. Click on the link then the *Install* button to add it.

1 Unforced-undamped vibrating system

Download POAV_Simscape_Mass_Spring_Damper.slx from Blackboard and open it in Matlab. You should be presented with a block diagram like the one below, plus a few other windows.



The green components and connections are the mechanical model. We have a mass and a spring connected to a position reference. There is also a damper, but this is commented out so inactive.

Double click on the mass and spring blocks; these will open in new windows. You'll see that each has a single parameter, mass or spring rate, each with the expected units. These are currently set to "m" and "k" respectively, rather than numeric values, which means Simulink will look for variables with those names in the Matlab workspace and read their values into the parameters for these blocks.

This means you will need to define "m" and "k" in the Matlab workspace before this Simulink model will run. We will start with a 3.6 kg mass and a spring of stiffness 8000N/m (you can change these later if you like). So type "`m = 3.6;`" and "`k = 8000;`" in the Matlab Command Window.

The role of the other blocks is as follows:

- The Solver Configuration block tells Simulink to solve the Simscape system.
- The Position Sensor extracts the position of the mass/spring connection, relative to a fixed Position Reference, and outputs it as a signal.
- The Scope shows a time history plot like an oscilloscope.
- The Convert to sampled data block converts the default Simulink signal type into sampled data, suitable for the Spectrum Analyser.
- There is also a Physical System to Simulink converter on the input to the Analysis Area.

Double click on any of these to see its settings. For example, the settings for the Spectrum Analyser are quite similar to the parameters of the Matlab function `pwelch`.

Tasks for you:

a) Run the model by clicking the green *Run* button, then take a look at the plots produced by the Scope and the Spectrum Analyser. Can you explain what you see? Note that the spring includes an initial displacement of 0.1m, so the system is responding to being released from a state of initial compression.
b) Calculate the natural frequency of the undamped mass-spring system using the formula from Kinsler and Frey section 1.2. How does this compare to the displacement spectrum peak amplitude found by the Spectrum Analyser? Can you explain any differences and how might you improve the numerical estimate of the natural frequency?
c) Experiment with increasing and decreasing the mass and stiffness of this system and observe the change in resonant frequency

2 Unforced-damped vibrating system

Uncomment the Damper block (right click on it then click on *Uncomment*) then double click on it to reveal its settings. You'll see that it is set to read the damping coefficient from a workspace variable called "R". Create this and set it to the value that will yield a critically damped system (see slides or Kinsler & Frey section 1.6).

Tasks for you:

d) Run the model then look at the plot produced by the Scope. Can you explain what you see?
e) Try reducing and increase the value of R. Can you explain the changes it causes?

3 Forced Damped vibrating system

We will now consider the response of the system to a driving force. To do this, add these blocks:

- An ideal force source (Simscape Foundation Library);
- A Colored Noise source (DSP Systems Toolbox);
- A Simulink to PS (Physical Signal) convertor.

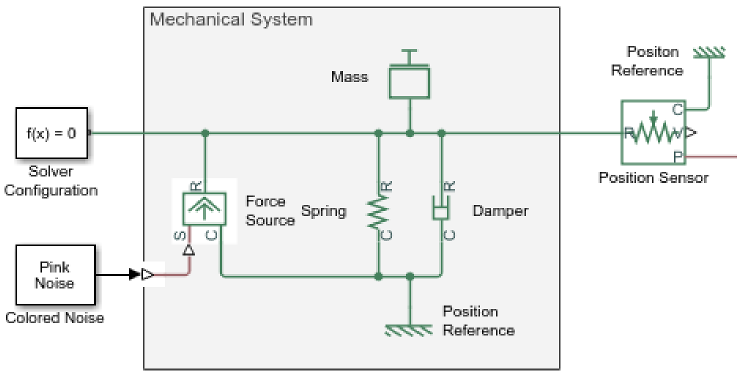
To add a block, either double click on the canvas and start typing its name or open the Library Browser and search for it there. Connect them as shown below and extend the Stop Time to 100s.

Double click on the Coloured Noise block. Set *Noise Color* to *white*, *Number of samples per output channel* to be 1 and *Output sample time* to be 1e-3s, to match the *Convert to sampled data* block.

On the Spectrum Analyser, increase the number of averages to be 100. In the workspace, set the damping coefficient R to 10 N/(m/s). You may also wish to try a stiffer spring e.g. `k = 20000;`

Tasks for you:

f) Observe where the resonance is located within the Spectrum Analyser; does this correspond with the expected resonant frequency?
g) Change the damping and observe the change in the response.



4 Extension Task

- Replace the Colored Noise source with sinusoidal excitation.
- Use an *RMS* block and a *Display* block to respectively compute the RMS and display its value.
- For a range of frequencies, you choose, compare the displacement RMS amplitude to the value predicted using the formulae in Kinsler & Frey section 1.7.

Hint: You can use the *To Workspace* block to return your RMS displacement amplitude to the workspace, where you can compute the error between it and the version you compute direct from the transfer function given in Kinsler & Frey.

5 Extended Extension Task for Matlab Enthusiasts!

- Create a for loop in Matlab that loops over an array of test frequencies you choose.
- For each of these, set the source frequency as above, then use `sim('POAV_Simscape_Mass_Spring_Damper')` to run the Simulink model.
- Collect the RMS displacement results in an array, then plot this versus frequency and compare to the theoretical transfer function from Kinsler and Frey.